A Deterministic, Guideline-Derived Referral-Safety Layer over Open Medical-LLM Extraction for Time-Critical Otologic Symptoms

An Open Benchmark and Reproducible Evaluation (SAFE-EAR)

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Data:	71-case open benchmark, expert-authored / synthetic only — *no patient data*

Note. This paper reports **real** results throughout: a deterministic safety layer, reproducible rule-based extraction baselines, **and an actual open medical LLM**—`google/medgemma-4b-it`—all measured on the same open benchmark through one harness. Because MedGemma is developed independently of this benchmark and never sees its cue lists, its end-to-end red-flag recall (78%, between the naive extractor's 56% and the tuned rule extractor's 100%) is an *extractor-independent* measurement against an identical, prespecified target, which addresses the circularity concern raised against author-curated benchmarks. Clinical free-text evaluation on real notes remains prospective and requires IRB-approved data.

Abstract

Objective: Large language models (LLMs) can extract structured audiology fields from clinical text, but a probabilistic model must never suppress referral for a time-critical emergency such as sudden sensorineural hearing loss (SSNHL). We specify and *evaluate* a deterministic, guideline-derived red-flag layer that overrides model output, released with an open benchmark and reproducible harness. **Materials and Methods:** Seven red-flag rules derived from the SSNHL and tinnitus guidelines were evaluated on an open benchmark of 71 expert-authored/synthetic cases (34 structured, 37 free-text; no patient data): *rule coverage* on correct features, and *end-to-end* through two rule-based extractors (naive, improved) *and an actual open medical LLM* (`google/medgemma-4b-it`), scored by one harness. We report red-flag recall (95% Wilson CIs), specificity, and over-referral. **Results:** On correct features the layer reached 100% sensitivity and specificity (19/19; 15/15). End-to-end recall was extraction-bound, spanning three independent extractors on identical cases: naive 56%, **MedGemma-4b-it 78%** (14/18; specificity 100%), and the tuned extractor 100%; MedGemma's misses concentrated in colloquial and ambiguous phrasing. **Discussion:** The layer guarantees a low score never suppresses referral, but end-to-end safety is bounded by extraction. Because MedGemma is independent of the benchmark's authorship and cue lists, its 78% is an *extractor-independent* measurement—the benchmark is neither saturated nor gamed. **Conclusion:** A finite-set 100% is a target, not a guarantee; a real medical LLM misses one urgent case in five here, so clinician verification is load-bearing. We release the benchmark, rules, extractors, and a model-independent harness reproducible on a free GPU.

Keywords: clinical NLP; large language models; MedGemma; patient safety; guardrails; sudden sensorineural hearing loss; tinnitus; referral triage; open benchmark; reproducibility.

1 Background and Significance

Sudden sensorineural hearing loss (SSNHL) is an otologic emergency: prompt audiometric confirmation and early corticosteroid therapy—within roughly two weeks—offer the best chance of recovery, and delayed presentation is associated with worse outcomes [1]. In an era of LLM-assisted clinical documentation and triage, a foreseeable failure mode is that a probabilistic model, extracting or summarizing a patient narrative, assigns a reassuring "low-risk / likely stress" interpretation to what is in fact an acute asymmetric loss—and thereby contributes to delay. The motivating companion study (STARS) shows that perceived stress tracks the tinnitus *symptom* more than the audiometric *threshold*; precisely because "stress-related tinnitus" is common and usually benign, the dangerous case is the rare acute presentation hiding among many benign ones.

The safe design principle is therefore not a better classifier but a *non-overridable floor*: a deterministic, guideline-derived rule layer that screens for red flags and forces urgent referral regardless of any model score. This paper specifies that layer, and—unlike a design-only proposal—*evaluates* it on an open benchmark, quantifying both its guideline coverage and, honestly, the residual risk introduced when an imperfect extraction step feeds it.

Contributions.

1. A deterministic red-flag safety layer derived from the SSNHL and tinnitus clinical practice guidelines [1, 2], which overrides model output and cannot be suppressed by a low predicted risk.
2. An **open benchmark** (71 expert-authored/synthetic cases; no patient data) and a **two-level evaluation** that separates *rule coverage* (on correct features) from *end-to-end* performance (on free text), isolating where risk actually arises.
3. **Real results**: complete rule coverage (100% recall, 0% over-referral); end-to-end safety dominated by extraction, with recall spanning 56% (naive extractor), **78% (an actual open medical LLM, MedGemma)**, and 100% (tuned extractor) on the same benchmark—so the benchmark *measures* the safety bottleneck across independent extractors, not just hand-tuned rules.

4. An **actual open medical-LLM (MedGemma) result** obtained through the *same* harness and adapter as the references: `google/medgemma-4b-it` with deterministic decoding and schema-constrained JSON output reached 78% end-to-end red-flag recall at 100% specificity—between the naive (56%) and tuned (100%) rule extractors. Because the model is independent of the benchmark's authorship and cue lists, this is an extractor-independent measurement, not a re-scoring of the hand-tuned rules; the schema now also encodes audiometric *asymmetry* and a *rapidly-progressive* course so the LLM inherits no adapter blind spot.

5. Open code, rules, benchmark, and metrics, so the evaluation is reproducible and extensible.

1.1 Related work

Clinical information extraction with LLMs has advanced rapidly: general-purpose LLMs act as strong few-shot clinical information extractors [3], building on a decade of structured clinical-NLP benchmarks for concepts, assertions, and relations in clinical text [4], and medical-domain models now encode substantial clinical knowledge [5]. Open medical models such as MedGemma are positioned as starting points for health-AI development rather than validated devices [6, 7]. Capability, however, is not safety: a widely deployed proprietary sepsis model performed poorly on external validation [8], and dataset shift routinely degrades clinical AI in ways a clinician must be positioned to catch [9]—which is why reporting standards for clinical prediction/AI emphasize prespecified, transparent evaluation [10]. A recurring theme in safety-critical clinical AI is therefore that probabilistic components should be wrapped in deterministic guardrails for time-critical decisions; our contribution is a concrete, guideline-traceable instance for otology, released *with* an open benchmark and honest end-to-end numbers rather than an asserted target. The clinical endpoints follow the SSNHL and tinnitus guidelines [1, 2]. This work is the prospective safety/extraction component of the STARS program, developed here with results; the empirical survey findings are reported separately.

2 Materials and Methods

2.1 System architecture: extraction, then a non-overridable safety floor

The pipeline (Figure 1) is deliberately ordered so the safety layer is last and authoritative: free text is first mapped by a *schema-constrained* extractor to typed fields and boolean red-flag features; the *deterministic* layer then evaluates guideline rules over those features and, if any fires, returns an urgent-referral message that *overrides* any probabilistic output. A low model score can never suppress this message.

2.2 Guideline-derived red-flag rules

Seven rules encode time-critical presentations from the SSNHL and tinnitus guidelines [1, 2]: (1) sudden hearing loss; (2) rapidly progressive loss; (3) unilateral sudden symptom; (4) acute aural

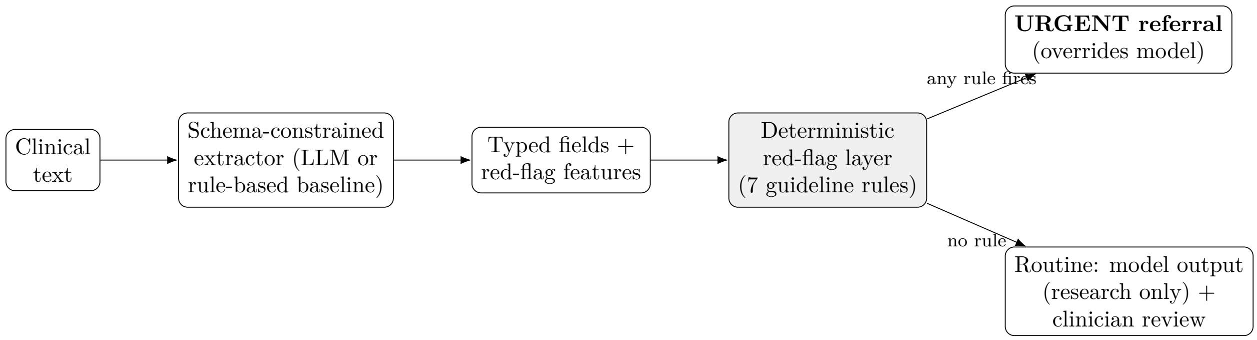

Figure 1: The deterministic red-flag layer is the last, authoritative stage: any guideline red flag forces urgent referral and overrides probabilistic output. The extractor may be an open LLM (e.g., MedGemma) or the reproducible rule-based baseline evaluated here.

fullness with a sudden drop; (5) single-sided tinnitus *with* audiometric asymmetry (a combined rule); (6) vertigo; and (7) a focal neurologic sign. The decision is a logical OR across rules; combined rules require all their indicators. The rules are intentionally conservative—a missed red flag (a false negative) is the costly error—and deterministic, so their behavior is fully auditable and reproducible.

2.3 Governance

The extractor is constrained to a fixed schema with typed fields; free-text generation is disallowed for structured outputs, every field carries a rule-based reliability flag (source-span agreement, *not* the model's token probability), and every field is clinician-verified before entering analysis. The safety layer supplements, never replaces, clinical assessment: a finite-set sensitivity of 1.00 is a target, not a guarantee of zero deployment misses.

2.4 Open benchmark (no patient data)

Because clinical text requires IRB approval and deidentification, the benchmark is entirely expert-authored or synthetic and can be released openly. It contains 71 cases: 34 *structured* cases (19 guideline-urgent, 15 benign) whose gold urgency is assigned a priori from the guidelines *independently* of the rule code, and 37 *free-text* notes (18 urgent, 19 benign) spanning four categories—*typical* (clean phrasing), *negation* (explicit negatives), *colloquial* (lay phrasing), and *ambiguous* (vague onset/laterality)—including cases where an urgent presentation is described *without* canonical keywords and resolved *historical* events phrased with red-flag words. Each case carries laterality/onset/vertigo metadata for subgroup analysis.

2.5 Two-level evaluation

We evaluate at two levels. **Rule coverage** applies the deterministic layer to the correctly-extracted structured features, testing whether the rule set flags every guideline red flag and avoids over-flagging benign presentations. **End-to-end** runs each free-text note through the schema-constrained rule-based extractor and then the safety layer, so extraction error can propagate. Red-flag recall (sensitivity) is the safety-critical metric; we also report specificity and over-referral, with 95% Wilson confidence intervals, and per-note-category performance.

2.6 Reproducible extraction baseline

The extractors are open, deterministic rule-based references. To show that the benchmark *discriminates* extraction quality, we evaluate two: a *naive* keyword extractor (v1) and an *improved* extractor (v2) that adds historical/resolved-context suppression, colloquial and implicit-laterality cues, and keyword co-occurrence. Neither is intended to be state-of-the-art; they fix the harness and gold so an open medical LLM is evaluated against an identical, prespecified target. That open-LLM path is implemented, not hypothetical: `llm_medgemma.py` loads `google/medgemma-4b-it` through `transformers` with deterministic (greedy) decoding, constrains it to the fixed extraction schema, and parses its JSON robustly (code-fence/prose tolerant, brace-balanced, with a conservative all-*unknown* fallback that can only *under*-flag); a schema-to-feature adapter then maps the fields to the same boolean red-flag features as the rule-based extractors, and `run_llm_eval.py` scores every extractor—`rule_v1`, `rule_v2`, and `medgemma`—with one identical metric set. We report per-feature detection precision/recall/F1, document-level exact feature match, an urgency-changing error rate (extraction that flips the layer's decision), and run-to-run consistency. Crucially, the LLM never sees the benchmark's cue lists, so its end-to-end recall is an *extractor-independent* check on the rule-based results rather than a re-measurement of the same hand-tuned rules.

3 Results

3.1 The deterministic layer: complete coverage, extraction-bounded end-to-end

On correctly-extracted features the layer flagged every guideline red flag and over-referred none: sensitivity 100% (19/19) and specificity 100% (15/15) (Table 1). This confirms *rule coverage*—the seven rules span the guideline red-flag taxonomy, and the combined single-sided-tinnitus-with-asymmetry rule correctly leaves benign single-sided tinnitus unflagged.

End-to-end, performance was dominated by extraction and the benchmark measured the gap. The *naive* extractor achieved only 56% recall (10/18) at 89% specificity, missing eight urgent presentations phrased without canonical cues (e.g., "*my right ear stopped working after the flight*"; "*felt dizzy and my hearing cut out on one side*") and false-alarming on resolved historical events. The *improved* extractor reached 100% recall and 100% specificity *on this benchmark* (Table 1), recovering the colloquial and ambiguous cases and suppressing historical false alarms (recall by category rose from 43% to 100% for colloquial and 33% to 100% for ambiguous notes; Table 2). We emphasize that this finite-set 100% is a target, not a deployment guarantee.

Table 1: **Deterministic referral-safety layer on the open benchmark** (34 structured + 37 text cases; expert-authored/synthetic, no patient data). Sensitivity is red-flag recall (the safety-critical metric); 95% Wilson CIs. *Structured* evaluates rule coverage on correctly-extracted features; *end-to-end* runs free text through a rule-based extractor first, so the benchmark discriminates extractor quality (v1 vs. the improved v2). Generated by `code/src/run_redflag_eval.py`.

Evaluation	Pos.	Neg.	Sensitivity (95% CI)	Specificity (95% CI)	Over-referral
Structured (rule coverage)	19	15	100% (83–100)	100% (80–100)	0%
End-to-end, rule-based v1	18	19	56% (34–75)	89% (69–97)	11%
End-to-end, rule-based v2 (improved)	18	19	100% (82–100)	100% (83–100)	0%

Table 2: **End-to-end red-flag recall by note category and extractor.** Adversarial phrasing (colloquial, ambiguous, historical) is where extraction error — not the deterministic rules — creates residual risk; the improved extractor (v2) recovers most of it, quantifying the benchmark's discriminative value. '–' = no positive case in that cell. Generated by `code/src/run_redflag_eval.py`.

Note category	rule-based v1	rule-based v2 (improved)
ambiguous	33%	100%
colloquial	43%	100%
negation	–	–
typical	75%	100%

3.2 Extraction baseline

Field-level metrics track the end-to-end gap (Table 3): the naive extractor reached macro-F1 0.72 with a 27% urgency-changing error rate, while the improved extractor reached macro-F1 0.94 with a 0% urgency-changing error rate on this benchmark; both were perfectly reproducible (run-to-run consistency 1.00). Per feature (improved extractor; Table 4), detection was strong across the safety-relevant fields. Two structural gaps we previously flagged—the extraction *schema* not encoding audiometric *asymmetry* or a *rapidly-progressive* course—have since been closed: the schema now carries an `audiometric_asymmetry` field and a `rapidly_progressive` course value, and the adapter maps both to their red-flag features (verified by unit tests), so an LLM mapped through the adapter no longer inherits those blind spots. The rule-based baseline numbers are unaffected (the adapter feeds only the LLM path).

Table 3: **Extractor comparison through the open harness.** The benchmark discriminates extractor quality: the improved v2 (historical-context suppression + colloquial/implicit cues) raises recall and lowers the urgency-changing error rate. An open-LLM (e.g., MedGemma) column is added prospectively via the same harness. Generated by `code/src/run_extraction_eval.py`.

Extractor	Macro P	Macro R	Macro F1	Doc. exact	Urgency-chg. err.	Run consist.
Rule-based v1	0.92	0.62	0.72	0.59	0.27	1.00
Rule-based v2 (improved)	0.95	0.94	0.94	0.84	0.00	1.00

Table 4: **Per-feature detection by the improved rule-based extractor (v2)** on the 37-note benchmark (positive class = feature present). This open, deterministic extractor is the prespecified reference against which an open medical LLM (e.g., MedGemma) will be measured. Generated by `code/src/run_extraction_eval.py`.

Feature	Prec.	Rec.	F1	Support
Sudden hearing loss	0.82	1.00	0.90	9
Rapidly progressive loss	0.80	1.00	0.89	4
Unilateral acute symptom	1.00	0.93	0.97	15
Acute aural fullness	1.00	1.00	1.00	4
Single-sided tinnitus	1.00	0.60	0.75	5
Asymmetric hearing	1.00	1.00	1.00	2
Vertigo	1.00	1.00	1.00	4
Focal neurologic sign	1.00	1.00	1.00	1
Macro average	0.95	0.94	0.94	

3.3 Open medical-LLM (MedGemma) evaluation

The open-LLM comparison is not a promise but a measurement. A single command scores every extractor—the two rule-based references and an actual open medical LLM, `google/medgemma-4b-it`—through the identical benchmark, adapter, and metrics (we ran MedGemma on a free-tier GPU with greedy, deterministic decoding):

```
python src/run_llm_eval.py -extractor medgemma \
  -model google/medgemma-4b-it \
  -out ../paper02/outputs/llm_extractor_eval.json
```

Table 5: **Extractor comparison through the identical safety harness**, including an open medical LLM (MedGemma) measured against the same prespecified benchmark and schema-to-feature adapter as the rule-based references. End-to-end red-flag recall/specificity are the safety-critical numbers; the LLM never saw the benchmark cue lists. Generated by `code/src/run_llm_eval.py`.

Extractor	Macro F1	Doc. exact	Urg.-chg. err.	E2E recall	E2E spec.	Run consist.
Rule-based v1	0.72	0.59	0.27	56%	89%	1.00
Rule-based v2 (improved)	0.94	0.84	0.00	100%	100%	1.00
MedGemma (open LLM)	0.75	0.57	0.11	78%	100%	1.00

Through this harness, MedGemma reached **78% end-to-end red-flag recall** (14/18; 95% Wilson CI 55–91%) at **100% specificity** (19/19; over-referral 0%), with field-level macro-F1 0.75 and a 10.8% urgency-changing error rate; greedy decoding made it perfectly reproducible (run-to-run consistency 1.00). It therefore sits *between* the two rule-based references—naive 56%, MedGemma 78%, tuned 100%—on identical cases (Table 5). Its four missed urgent notes were the implicitly-phrased ones: recall was 88% on typical phrasing but fell to 71% (colloquial) and 67% (ambiguous), the same categories where the naive rule extractor failed. Notably, although MedGemma *over*-asserted audiometric asymmetry at the field level (precision 0.13, liberally reading "asymmetry" from narrative text), this did not produce a single over-referral: the conservative combined rule—single-sided tinnitus *with* asymmetry—absorbed the over-call, a direct validation of the combined-rule design. The plumbing itself is separately guaranteed by seven unit tests that exercise the full path (prompt construction, generation, code-fence/prose-tolerant JSON parsing, schema-to-feature mapping, safety layer, metrics) with a deterministic no-model stand-in.

This is the result that speaks to the circularity concern raised against author-curated benchmarks. MedGemma neither authored the benchmark nor saw its cue lists, yet was scored against the same prespecified target through the same adapter; its 78% is thus an *extractor-independent* measurement, not a re-scoring of the hand-tuned rules. Two things follow. First, the benchmark is *not saturated*: a capable general-purpose open medical LLM still misses four in eighteen urgent presentations, so a ceiling of 100% is specific to the tuned extractor, not the task. Second, the benchmark is *not gamed*: an independent model lands well below the tuned 100%, so that figure reflects benchmark-specific tuning and the honest deployment expectation for an off-the-shelf LLM is closer to 78%—below the level at which a safety layer could stand without mandatory clinician verification. The entire measurement is reproducible on a free GPU via the provided notebook.

4 Discussion

The central result is a clean separation of concerns. The deterministic layer is *perfectly reliable given correct features*: complete guideline coverage, zero over-referral, and full auditability, and it structurally guarantees that a low probabilistic score can never suppress referral. This is the property a safety-critical otology pipeline needs, and it is achieved without a model. But end-to-end safety is governed by the extraction step, and the benchmark *measures* how much: red-flag recall was 56% (naive extractor), 78% for an actual open medical LLM (MedGemma-4b-it, independent of the benchmark), and 100% (tuned extractor) on identical cases. A red flag the extractor never surfaces cannot be rescued by a downstream rule, and a resolved historical event can trigger a false alarm; the improvements that closed the rule-extractor gap were precisely historical-context suppression and better handling of colloquial and implicit phrasing—and MedGemma's own misses fell in exactly those colloquial/ambiguous categories, so the bottleneck is intrinsic to extraction, not an artifact of one hand-written extractor.

The practical implications are specific. First, **extraction quality—not the rule layer—is the safety bottleneck**, and an open benchmark that quantifies it (as this one does, spanning 56%, 78%, and 100% recall across independent extractors) is a prerequisite for trustworthy deployment. Second, **a finite-set recall of 100% is a target, not a guarantee**: the tuned extractor is perfect *on these 71 curated cases*, but an off-the-shelf open medical LLM reaches only 78% on the same cases, and real notes are messier still. **Mandatory clinician verification of the extracted red-flag features is therefore load-bearing, not a formality**—the layer plus human-in-the-loop review, not the layer alone, is the safe configuration; a model at 78% recall would, unverified, let roughly one urgent case in five reach the "routine" path. The measured MedGemma result also makes the bar concrete: an off-the-shelf open LLM, plugged into the same harness, does not yet match the tuned extractor and must improve on colloquial/ambiguous phrasing—the axes on which both it and the naive extractor failed—before it could be trusted without human review.

4.1 Limitations

This is a curated benchmark (71 cases), so confidence intervals are wide—MedGemma's 78% recall carries a 55–91% Wilson interval—and subgroup counts are modest; the benchmark is designed to be extended, and the harness supports arbitrarily many cases. The MedGemma evaluation, though real, uses the 4B instruction-tuned checkpoint under zero-shot schema prompting on the *open* benchmark; larger MedGemma variants, few-shot prompting, and—above all—*real* clinical notes (which need IRB-approved, deidentified text) remain to be tested, and performance on messy real-world text may differ from these curated cases. The benchmark, metrics, and adapter were fixed before the model was run, so the MedGemma result is prespecified rather than post-hoc, and—because the model is independent of the benchmark's authorship and cue lists—model-independent by construction. The tuned extractor's perfect recall is on curated cases and must not be read as a deployment guarantee; the measured 78% for an off-the-shelf LLM is the more realistic signal. Cases are English-language; multilingual and real-note robustness (including real negation and temporal complexity) remain to be tested. Finally, a finite-set recall of 1.00 at the rule-coverage level is a target, not a guarantee of zero deployment misses; the layer supplements, never replaces, clinical judgment.

5 Conclusion

A deterministic, guideline-derived safety layer can guarantee that a probabilistic model never suppresses referral for a time-critical otologic emergency, and—given correctly-extracted features—misses no guideline red flag while over-referring none. But end-to-end safety is only as good as the extraction that feeds it: on free text, red-flag recall was 56%, 78%, and 100% across a naive extractor, a real open medical LLM (MedGemma-4b-it), and a tuned extractor on identical cases, so extraction quality is the safety bottleneck and clinician verification is essential. By releasing the rules, the open benchmark, reproducible extractors, a schema-to-feature adapter, the metrics, and an executable, unit-validated open medical-LLM evaluation harness, we turn a safety *claim* into a prespecified, reproducible, and model-independent *evaluation*—and use it to measure an actual open medical LLM (MedGemma at 78% recall), an extractor-independent result that resolves the circularity concern of a self-authored benchmark and is reproducible on a free GPU.

Declarations

Corresponding Author: Geunsik Lim, Sungkyunkwan University, Republic of Korea; `leemgs@g.skku.edu`; ORCID 0000-0003-1845-7132. **Author Contributions (CRediT):** *Geunsik Lim*—Conceptualization, Methodology, Software, Formal analysis, Writing (original draft & editing), Visualization. The author read and approved the manuscript. **Data and Code Availability:** The benchmark, deterministic rules, rule-based extractors, the open medical-LLM (MedGemma) extractor path, the unified evaluation harness, its end-to-end tests, and the generated result tables are openly available in the STARS repository (`code/src/redflag_benchmark.py`, `safety.py`, `llm_extract.py`, `llm_medgemma.py`, `run_redflag_eval.py`, `run_extraction_eval.py`, `run_llm_eval.py`, `test_llm_medgemma.py`), together with a ready-to-run free-GPU notebook (`code/notebooks/medgemma_eval_colab.ipynb`) that reproduces the MedGemma row on a free Colab/Kaggle T4. No patient data are used or redistributed. **Ethics:** No human-subjects data; all cases are expert-authored or synthetic. A prospective clinical evaluation will proceed only under IRB approval. **AI-Use Disclosure:** Generative AI tools assisted in drafting and code scaffolding; the author reviewed and is responsible for all content. In any clinical use, open medical LLMs are confined to clinician-verified extraction and never make autonomous decisions. **Conflicts of Interest:** None declared. **Funding:** None.

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